

Primary Decomposition

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December 2024

1 Introduction

If we consider an element $a \in A$ UFD, then we have a unique decomposition $a = p_1^{e_1} \dots p_n^{e_n}$ up to permutation and multiplications with unit. Unfortunately, we can find a lot of ring that are not UFD. Let us start with an example:

Example 1.1. Let be R a PID, then for any non-zero I ideal of R , I can be written as $I = (p_1^{e_1} \dots p_n^{e_n})$ for some $p_i \in R$ prime and $k_i \in \mathbb{N}$. The "prettiest" deomposition of thi ideal is $I = (p_1^{k_1}) \cap \dots \cap (p_n^{k_n})$.

Our goal is to generalize the concept of prime factorization for rings that are not UFD. We will do this on ideal as the same as we have done in the example 1.1. For more generality, we are going to define al the concept on rings and not on ideal principle domain.

2 First attempt

Definition 2.1. Let R be a ring, an ideal $Q \subset R$ with $Q \neq R$ is colled primary if $\forall a, b \in R$ with $ab \in Q$ we have $a \in Q$ or $b^n \in Q$ for some $n \in \mathbb{N}$.

Remark 2.2. The second part of the definition is equivalent to ask $b \in \sqrt{Q}$.

Proposition 2.3. Let R be a PID, then the primary ideal are only the ideal I of the form $I = (p^n)$ for a $p \in R$ prime and $n \in \mathbb{N}$.

Proof. Let R be a ring:

- (p^n) is primary: if $ab \in (p^n)$ then $p^n | ab$. Now we have two chances: $p^n | a$ or $p | b$. In the first case $a \in (p^n) = I$, in the second case $b \in (p) \implies b^n \in (p^n) \implies b \in \sqrt{(p^n)}$;
- On the other hand, let $I = (p_1^{k_1} \dots p_n^{k_n})$ a primary ideal with p_i prime. Supposing $n > 1$ then $p_1^{k_1}(p_2^{k_2} \dots p_n^{k_n}) \in I$ but neither $p_2^{k_2} \dots p_n^{k_n}$ nor any power of $p_1^{k_1}$ are in I .

□

So the definition of primary ideal does not correspond to the one of prime ideal.

Corollary 2.4. Exists primary ideal that are not prime.

Example 2.5. $(2) \in \mathbb{Z}$ is prime and primary; (4) is primary for the last proposition but not prime because $2 \cdot 2 \in (4)$ but $2 \notin (4)$.

Lemma 2.6. If Q is primary, then $\sqrt{Q}$ is prime.

Proof. If $ab \in \sqrt{Q} \implies (ab)^n \in Q$ for some $n \in \mathbb{N}$. Hence $a^n \in Q$ or $b^n \in Q \implies (b^n)^m \in Q$ for some $m \in \mathbb{N}$, which means that a or b lie in $\sqrt{Q}$. $\square$

The lemma allows us to give the following definition:

Definition 2.7. Let Q be a primary ideal, then we say that Q is P -primary where $P = \sqrt{Q}$ is a prime ideal.

Now we will state three lemmas in order to give the proof of an interesting way to detect primary ideals:

Lemma 2.8. Let $J \subset I \subset R$ ideals:

1. J is maximal in $R \iff J/I$ is maximal in R/I .
2. $\sqrt{0}$ is a maximal $\iff$ Every element of R is either a unit or nilpotent.

Remark 2.9. The condition of an ideal $Q \subsetneq R$ being primary can also be expressed in terms of the quotient ring R/Q , so we can require that: if $\overline{ab} = 0 \implies \overline{a} = 0$ or $\overline{b}^n = 0$ for some $n \in \mathbb{N}$.

We also have a condition with the zero-divisors:

Lemma 2.10. Q is primary $\iff$ every zero-divisor of R/Q is nilpotent:

Proof. We will demonstrate the two directions:

- $\implies$: Let $x \in R/Q$ zero-divisor $\neq 0$ (to avoid trivial cases) $\implies \exists b \neq 0 \in R/Q$ such that $\overline{bx} = 0 \implies \overline{x}^n = 0$.
- $\impliedby$: Let $ab \in Q \implies \overline{ab} = 0$ then $\overline{a} = 0$ or b is zero-divisor, so $\overline{b}^n = 0$ for some n .

$\square$

Proposition 2.11. Let P be a maximal ideal in a ring R . If an ideal $Q \subset R$ is such that $\sqrt{Q} = P$, then Q is P -primary.

Proof. As $\sqrt{Q}$ is maximal, lemma 2.8 (1) $\implies \sqrt{0}$ is maximal in R/Q , by lemma 2.8 (2) $\implies$ every element of R/Q is a unit or nilpotent, so every zero-divisor (which is never a unit) is nilpotent. By proposition 2.10, Q is primary. $\square$

Now we can give the formal definition for decomposition:

$$\overline{b} = 0$$

Definition 2.12. Let I be an ideal in a ring R . A primary decomposition of I is a finite set $\{Q_1, \dots, Q_n\}$ of primary ideals such that $I = Q_1 \cap \dots \cap Q_n$

Example 2.13. In the example 1.1 $\{(p_1^{k_1}), \dots, (p_n^{k_n})\}$ is a primary decomposition.

At this point, it would be great having a unique decomposition for the ideal like the one we have for element in UFD. Unfortunately our definition is too "weak" to get the uniqueness; there are two most reason for that:

Example 2.14. Let's see these two examples:

1. "Superfluous ideals": $(x^2) = (x^2) \cap (x) \in \mathbb{R}[x]$ are two different decomposition of the same ideal (x^2) , in this case (x) is "superfluous";
2. "Same radical": $(x^2, xy, y^2) = (x^2, y) \cap (x, y^2) \in \mathbb{R}[x, y]$, none of the ideals is "superfluous". All three ideals have the same radical (x, y) , that is maximal, so are all primary. So we have also two decomposition in which none of the ideals is superfluous.

3 Second attempt

In this section we are going to solve the issues that came up in the upper lines.

Definition 3.1. Let $\{Q_1, \dots, Q_n\}$ be a primary decomposition of an ideal I and let $P_i := \sqrt{Q_i}$ for $i \in \{1, \dots, n\}$. Then the decomposition is called minimal if:

1. $\forall i \in \{1, \dots, n\}, \bigcap_{j \neq i} Q_j \not\subseteq Q_i$;
2. $P_i \neq P_j \forall i \neq j$.

Lemma 3.2. Let P a prime ideal in a ring R , if Q_1 and Q_2 are two P -primary ideals in R , then $Q_1 \cap Q_2$ is P -primary as well.

Proof. Let $ab \in Q_1 \cap Q_2 \implies \{(a \in Q_1 \text{ or } b \in P) \text{ and } (a \in Q_2 \text{ or } b \in P)\} \implies a \in Q_1 \cap Q_2 \text{ or } b \in P$. We also have that $\sqrt{Q_1 \cap Q_2} = \sqrt{Q_1} \cap \sqrt{Q_2} = P \cap P = P$, so potentially $b \in P \implies b \in \sqrt{Q_1 \cap Q_2}$. $\square$

Corollary 3.3. If an ideal has a primary decomposition, it also has a minimal one.

Proof. Let $\{Q_1, \dots, Q_n\}$ a primary decomposition of an ideal $I \subset R$, with R a ring. Now for all $i \in \{1, \dots, n\}$:

- if the first condition of the definition 3.1 is not satisfied pull out Q_i from the decomposition;
- if the second condition of the definition 3.1 is not satisfied replace Q_i and Q_j with $Q_i \cap Q_j$.

²As Q_1 and Q_2 are P -primary we have that.

□

Example 3.4. Let us consider the ideal $I := (y) \cdot (x, y)$. I has two different primary decomposition:

1. $I = Q_1 \cap Q_2 = (y) \cap (x^2, xy, y^2)$;
2. $I = Q_1 \cap Q'_2 = (y) \cap (x, y^2)$.

Q_1 is primary because is prime, and Q_2 and Q'_2 are primary by proposition 2.11 since $\sqrt{Q_2} = \sqrt{Q'_2} = (x, y)$ is maximal. We can observe that none of the ideals is superfluous, and that $\sqrt{Q_1} = Q_1 \neq (x, y)$. Then 1 and 2 are minimal primary decomposition.

So the decomposition is not unique, again. Geometrically wise, could happens that a subvariety is contained in another subvariety, i.e. $Z(P_i) \subsetneq Z(P_j)$ for some $i, j \in \{1, \dots, n\}$ with Q_i and Q_j primary ideal. Obviously, the corresponding algebraic statement is that $P_j \subsetneq P_i$, and we say that P_j is an embedded isolated prime ideal.

4 Uniqueness theorems

In this section we will finally have two uniqueness theorem, but we have to work a little.

Lemma 4.1. Let Q be a P -primary ideal in a ring R . Then $\forall a \in R$:

$$\sqrt{Q : a} = \begin{cases} R & \text{if } a \in Q \\ P & \text{if } a \notin Q \end{cases}$$

Proof. If $a \in Q \implies Q : a = R$ and $\sqrt{Q : a} = R$ as well. If $a \notin Q$ for any $b \in (Q : a)$, $ba \in Q$ so $b \in \sqrt{Q} = P$ since Q is P -primary. Hence $Q \subset Q : a \subset P$, and by taking the radicals we obtain $P = \sqrt{Q} \subset \sqrt{Q : a} \subset P$. □

Definition 4.2. Let I be an ideal in a ring R :

1. An associated prime ideal of I is a prime ideal that can be written as $\sqrt{I : a}$ for some $a \in R$. And we denote by $Ass(I)$ the set of these ideal.
2. The minimal elements of $Ass(I)$ are called isolated prime ideals, the other ones embedded prime ideals of I .

Lemma 4.3. Let $I = \bigcap_{i=1}^n Q_i \implies \forall a \in R, (I : a) = \bigcap_{i=1}^n (Q_i : a)$.

Lemma 4.4. Let $I_1, \dots, I_n \subset R$ ideals and $P \subset R$ prime ideal such that $P \supset I_1 \cap \dots \cap I_n \implies \exists k \in \{1, \dots, n\}$ tale che $P \supset I_k$.

Finally we can give the first uniqueness theorem.

Theorem 4.5. Let $Q = \{Q_1, \dots, Q_n\}$ a minimal primary decomposition for an ideal I and let $P_i = \sqrt{Q_i}$, then $\{P_1, \dots, P_n\} = Ass(I)$.

Proof. Let's demonstrate the double inclusion:

- " $\subset$ ": Let be $P_i \in Q$, as the decomposition is minimal $\bigcap_{j \neq i} Q_j \not\subset P_i$, so we can find $a \in \bigcap_{j \neq i} Q_j$ with $a \notin P_i$. Then:

$$\begin{aligned} \sqrt{I : a} &= \sqrt{(Q_1 : a) \cap \dots \cap (Q_n : a)} = \sqrt{Q_1 : a} \cap \dots \cap \sqrt{Q_n : a} \\ &\stackrel{4}{=} R \cap \dots \cap P_i \cap \dots \cap R = P_i \end{aligned}$$

- " $\supset$ ": Let $P \in \text{Ass}(I)$, so $P = \sqrt{I : a}$ for some $a \in R$. Then as above we have

$$P = \sqrt{I : a} = \sqrt{Q_1 : a} \cap \dots \cap \sqrt{Q_n : a}$$

end by lemma 4.4 $P \supset \sqrt{Q_i : a}$ for some $i \in \{1, \dots, n\}$ since P is prime. The above equation also implies $P \subset \sqrt{Q_i : a}$ and so $P = \sqrt{Q_i : a}$. By lemma 4.1 we have that P can be P_i or R , and since P is prime we conclude that $P = P_i$. $\square$

Corollary 4.6. First uniqueness theorem. The number of components in a minimal primary decomposition and their radicals do not depend on the chosen decomposition.

Theorem 4.7. Second uniqueness theorem. In a minimal primary decomposition the primary components corresponding to minimal prime ideals do not depend on the chosen decomposition.

So we do not have the "absolute uniqueness", but for all the decomposition the number of primary component and the primary components corresponding to minimal prime ideals are universal are the same, so they are "universal".

5 In a Noetherian ring

If R is more then a "simple" ring, what can we say? If R is a Noetherian ring we have the following proposition.

Theorem 5.1. In a Noetherian ring every ideal has a primary decomposition.

Proof. Assume for contraddiction that R is a Noetherian ring that has an ideal without primary decomposition. We can consider the family F of the all ideals without a primary decomposition and because R is Noetherian exists an ideal $I \subset R$ that is maximal among F . We know by the correspondence that all the ideals in R/I are ideals in R that include I , so by the maximality of I we have that the only ideal in R/I without a decomposition is (0) ⁵. I does not admit a decomposition, so neither it is primary and so exists $ab \in I$ such that $ab = 0$ but $a \neq 0$ and $b^n \neq 0$ for all $n \in \mathbb{N}$. As R is Noetherian, also R/I is Noetherian and so the chain

$$\text{ann}(b) \subset \text{ann}(b^2) \subset \dots$$

³By lemma 4.3.

⁴By lemma 4.1.

⁵For an ideal $I \subset Q$ we have that Q is primary if and only if Q/I is primary in R/I .

becomes stationary, i.e. exists $n \in \mathbb{N}$ such that $\text{ann}(b^n) = \text{ann}(b^{n+1})$. As $(a) \neq (0) \neq (b^n)$ these two ideals have a primary decomposition, and by taking the primary decomposition of these two we obtain a primary decomposition of $(a) \cap (b^n)$. At this point, if $(a) \cap (b^n) = (0)$ we have done since we have found a primary decomposition for (0) :

- $\subset$: if $x \in (a) \cap (b^n)$ then $\exists c, d \in R/I$ such that $db^n = x = ca$. As $ab = 0$ we have $0 = cab = xb = db^{n+1}$, hence $d \in \text{ann}(b^{n+1}) = \text{ann}(b^n)$, and so $x = db^n = 0$
- $\supset$: always true.

□

Corollary 5.2. In a Noetherian ring every ideal has a minimal primary decomposition.

Proof. The statement follows by theorem 5.1 and by corollary 3.3

□